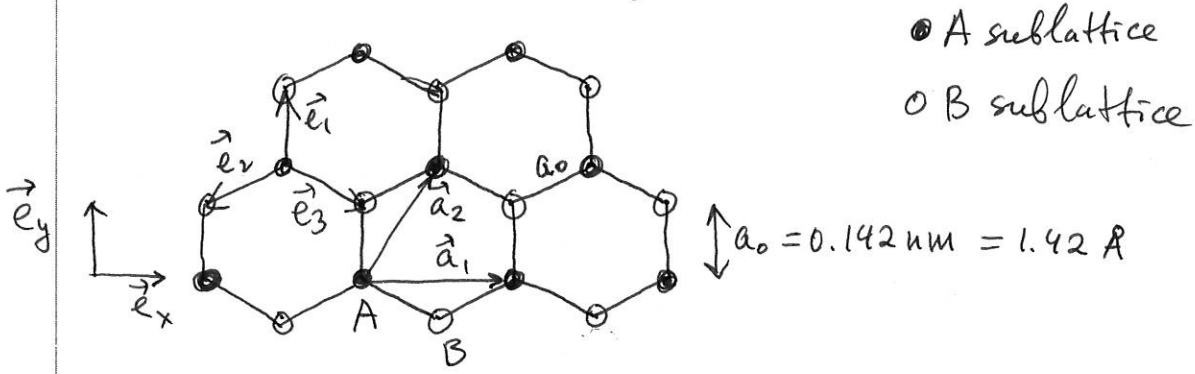
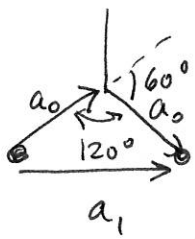
Graphene structure: honeycomb lattice



Honeycomb is not a Bravais lattice: sites A and B however form triangular Bravais lattice with basis vectors $\vec{a}_1, \vec{a}_2$.
Any site on A lattice: $\vec{R}_A = n_1 \vec{a}_1 + n_2 \vec{a}_2$.



$$|a_1|^2 = |a_0|^2 + |a_0|^2 + 2|a_0|^2 \underbrace{\cos\frac{2\pi}{3}}_{+1/2} = 3|a_0|^2$$

$$\Rightarrow a_2 = a_1 = \sqrt{3}\, a_0$$

Reciprocal lattice formed by $\vec{b}_1, \vec{b}_2$ such that

$$\vec{a}_i \cdot \vec{b}_j = 2\pi\, \delta_{ij}$$

Any vector $\vec{K}$ of reciprocal lattice $\vec{K} = m_1 \vec{b}_1 + m_2 \vec{b}_2$.
Hence $\vec{K} \cdot \vec{R}_A = (m_1 \vec{b}_1 + m_2 \vec{b}_2) \cdot (n_1 \vec{a}_1 + n_2 \vec{a}_2) =$

$$= m_1 n_1 \underbrace{\vec{b}_1 \cdot \vec{a}_1}_{2\pi} + m_1 n_2 \underbrace{\vec{b}_1 \cdot \vec{a}_2}_{0} + m_2 n_1 \underbrace{\vec{b}_2 \cdot \vec{a}_1}_{0}$$

$$+ m_2 n_2 \underbrace{\vec{b}_2 \cdot \vec{a}_2}_{2\pi} = 2\pi (m_1 n_1 + m_2 n_2)$$

$$\Rightarrow e^{i \vec{K} \cdot \vec{R}_A} = e^{i 2\pi (m_1 n_1 + m_2 n_2)} = 1.$$

A and B sites are connected by $(\vec{e}_1, \vec{e}_2, \vec{e}_3)$; $|\vec{e}_i| = e_i = a_0$

In terms of orthogonal basis $\vec{e}_x, \vec{e}_y$: $\vec{e}_1 = +a_0 \vec{e}_y$

$$\vec{e}_2 = -\frac{a_0}{2}\left(\sqrt{3}\,\vec{e}_x + \vec{e}_y\right)$$

$$\vec{e}_3 = \frac{a_0}{2}\left(\sqrt{3}\,\vec{e}_x - \vec{e}_y\right)$$

Triangular basis vectors are

$$\vec{a}_1 = \sqrt{3}\,a_0\,\vec{e}_x\,, \quad \vec{a}_2 = \frac{\sqrt{3}\,a_0}{2}\left(\vec{e}_x + \sqrt{3}\,\vec{e}_y\right)$$

so that $|\vec{a}_{1,2}| = \sqrt{3}\,a_0$.

General construction for $\vec{b}_i$:

$$\vec{b}_1 = 2\pi \frac{\vec{a}_2 \times \vec{a}_3}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)}$$

$$\vec{b}_2 = 2\pi \frac{\vec{a}_3 \times \vec{a}_1}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)}$$

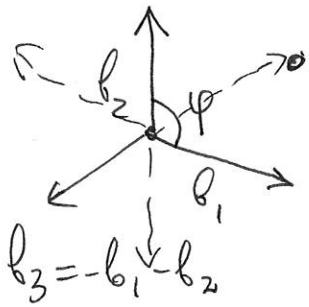
$$\vec{b}_3 = 2\pi \frac{\vec{a}_1 \times \vec{a}_2}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)}$$

Add $\vec{a}_3 = a_0 \vec{e}_z = a_0(0, 0, 1)$ so that triangular lattice repeats along z-axis

Volume $v = \vec{a}_1 \cdot \vec{a}_2 \times \vec{a}_3 = a_0^3 \begin{pmatrix} \sqrt{3} & 0 & 0 \\ \frac{\sqrt{3}}{2} & \frac{3}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} = \frac{3\sqrt{3}}{2} a_0^3$

$$\vec{b}_1 = \frac{2\pi}{\frac{3\sqrt{3}}{2} a_0^3} \cdot \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\sqrt{3}}{2} & \frac{3}{2} & 0 \\ 0 & 0 & 1 \end{vmatrix} a_0^2 = \frac{2\pi}{\sqrt{3}\,a_0}\left(\vec{e}_x - \frac{1}{\sqrt{3}}\,\vec{e}_y\right)$$

$$\vec{b}_2 = \frac{2\pi}{\frac{3\sqrt{3}}{2} a_0^3} \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ 0 & 0 & 1 \\ \sqrt{3} & 0 & 0 \end{vmatrix} a_0^2 = \frac{4\pi}{3a_0}\,\vec{e}_y$$



$$|\vec{b}_1| = \frac{2\pi}{\sqrt{3}a_0}\left(1+\frac{1}{3}\right)^{1/2} = \frac{2\pi}{\sqrt{3}a_0}\sqrt{\frac{4}{3}} = \frac{4\pi}{3a_0}$$

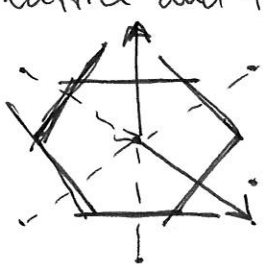
$$|\vec{b}_2| = \frac{4\pi}{3a_0}$$

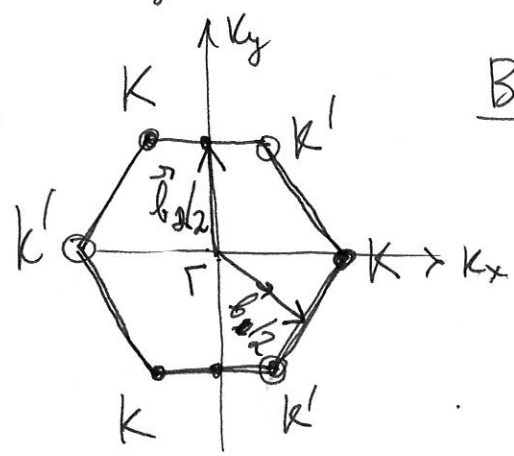
$$\vec{b}_1 \cdot \vec{b}_2 = \left(-\frac{2\pi}{3a_0}\vec{e}_y\right)\cdot\left(\frac{4\pi}{3a_0}\vec{e}_y\right) = +\left(\frac{4\pi}{3a_0}\right)^2 \underbrace{\cos\varphi}_{-1/2}$$

$\Rightarrow$ angle between $\vec{b}_1$ and $\vec{b}_2$ is $\frac{2\pi}{3}$.

Reciprocal lattice ~~is~~ triangular lattice too.

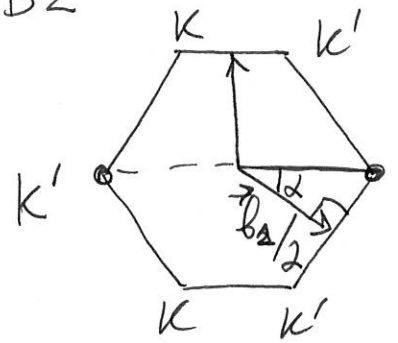
$\Rightarrow$ build reciprocal triang. lattice and then BZ:





Brillouin Zone

Corners of BZ



$$\vec{K} = \frac{4\pi\,\hat{x}}{3\sqrt{3}a_0} = \frac{4\pi}{3a}\,\hat{x}$$

$$\frac{1}{2}\vec{b}_1 \cdot \vec{e}_x = \frac{1}{2}\frac{2\pi}{\sqrt{3}a_0} = \frac{2\pi}{\sqrt{3}a_0}\cdot\frac{1}{2} \Rightarrow \cos\alpha = \frac{1}{2} \Rightarrow \alpha = \pi/3$$

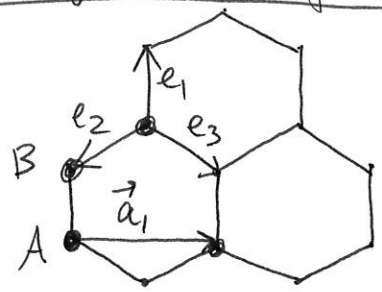
But $K \sin\alpha = \frac{b_1}{2} = \frac{2\pi}{3a_0}$

$$K \sin\frac{\pi}{3} = \frac{2\pi}{3a_0}, \quad K = \frac{4\pi}{3\sqrt{3}a_0}$$

$$\Rightarrow \vec{K} = \frac{4\pi}{3\sqrt{3}a_0}\vec{e}_x, \quad \vec{K}' = -\frac{4\pi}{3\sqrt{3}a_0}\vec{e}_x$$

All others can be obtained by reciprocal lattice translations.

Tight-binding model for graphene



In terms of $|\vec{a}_1| = \sqrt{3}\, a_0 \equiv a$, vectors connecting A and B sublattices are $(a_0 = a/\sqrt{3})$

$$\vec{e}_1 = \frac{a}{\sqrt{3}}\,\vec{e}_y = \frac{a}{\sqrt{3}}\,\hat{y}$$

$$\vec{e}_2 = -\frac{a}{2\sqrt{3}}\left(\sqrt{3}\,\vec{e}_x + \vec{e}_y\right) = a\left(-\frac{1}{2}\hat{x} - \frac{1}{2\sqrt{3}}\hat{y}\right)$$

$$\vec{e}_3 = \frac{a}{2\sqrt{3}}\left(\sqrt{3}\,\vec{e}_x - \vec{e}_y\right) = a\left(\frac{1}{2}\hat{x} - \frac{1}{2\sqrt{3}}\hat{y}\right)$$

let $c^{+}_{A/B,\vec{r},\sigma}$ create an electron with spin σ on the carbon p_z orbital on site $\vec{r}$ of A/B sublattice

$$H = -t \sum_{\langle rr' \rangle} \sum_{\sigma} \left(c^{+}_{A\vec{r}\sigma}\, c_{B\vec{r}'\sigma} + c^{+}_{B\vec{r}'\sigma}\, c_{A\vec{r}\sigma} \right)$$

sum over $\langle \vec{r}\,\vec{r}' \rangle$ = neighboring pairs of A-B sites

t = tunneling matrix element $= 2.97\,eV$
(determined by overlap between neighboring p_z orbitals)

$$H = -t \sum_{\sigma} \sum_{\vec{r} \in A} \sum_{j=1}^{3} \left(c^{+}_{A\vec{r}\sigma}\, c_{B\,\vec{r}+\vec{e}_j\,\sigma} + h.c. \right)$$

Transform to $\vec{k}$ space

$$c_{\alpha \vec{r} \sigma} = \frac{1}{\sqrt{N_A}} \sum_{\vec{k} \in BZ} e^{i \vec{k} \cdot \vec{r}} c_{\alpha \vec{k} \sigma}$$

$\alpha = A$ or B ; N_A = number of unit cells = $\frac{1}{2}$ Number of sites.

$$H = -t \sum_{\sigma} \sum_{k, k'} \sum_{j=1}^{3} \frac{1}{N_A} \sum_{\vec{r} \in A} \left(e^{-i \vec{k} \cdot \vec{r}} e^{i \vec{k}' (\vec{r} + \vec{e}_j)} c^{+}_{A \vec{k} \sigma} \cdot c_{B \vec{k}' \sigma} + h.c. \right)$$

$$\frac{1}{N_A} \sum_{\vec{r} \in A} e^{i \vec{r} \cdot (\vec{k}' - \vec{k})} = \delta_{\vec{k}, \vec{k}'}$$ (upto reciprocal lattice vector $\vec{G}$ which takes one of $\vec{k}$'s outside the BZ)

$$\Rightarrow H = -t \sum_{\vec{k} \in BZ} \sum_{j=1}^{3} \sum_{\sigma} \cdot \left(e^{i \vec{k} \cdot \vec{e}_j} c^{+}_{A k \sigma} c_{B k \sigma} + e^{-i \vec{k} \cdot \vec{e}_j} c^{+}_{B k \sigma} c_{A k \sigma} \right)$$

Note $$\sum_{j=1}^{3} e^{i \vec{k} \cdot \vec{e}_j} = e^{i \vec{k} \cdot \vec{e}_1} + e^{i \vec{k} \cdot \vec{e}_2} + e^{i \vec{k} \cdot \vec{e}_3} =$$

$$= e^{i k_y a / \sqrt{3}} + e^{-i k_x a/2 - i k_y a / 2\sqrt{3}} + e^{i k_x a/2 - i k_y a / 2\sqrt{3}}$$

$$= 2 \cos\left(\frac{k_x a}{2}\right) e^{-i \frac{k_y a}{2\sqrt{3}}} + e^{i \frac{k_y a}{\sqrt{3}}}$$

$$H = \sum_{\vec{k} \in BZ} \sum_{\sigma} \left(\xi(\vec{k}) c^{+}_{A \vec{k} \sigma} c_{B \vec{k} \sigma} + \xi^{*}(\vec{k}) c^{+}_{B \vec{k} \sigma} c_{A \vec{k} \sigma} \right)$$

$$\xi(\vec{k}) \equiv -t \sum_{j=1}^{3} e^{i \vec{k} \cdot \vec{e}_j}$$

Define $c_\sigma(\vec{k}) = \begin{pmatrix} c_{A\vec{k}\sigma} \\ c_{B\vec{k}\sigma} \end{pmatrix}$

$$c^\dagger_\sigma(\vec{k}) = \left(c^\dagger_{A\vec{k}\sigma},\ c^\dagger_{B\vec{k}\sigma}\right)$$

$$H = \sum_{\vec{k},\sigma} c^\dagger_\sigma(\vec{k}) \underbrace{\begin{pmatrix} 0 & \xi(k) \\ \xi^*(k) & 0 \end{pmatrix}}_{\mathcal{H}(\vec{k})} c_\sigma(k)$$

Eigenvalues of $\mathcal{H}(\vec{k})$

$$\det\left(\mathcal{H}(\vec{k}) - E(\vec{k})\, I\right) = 0 = \det\begin{pmatrix} -E & \xi(k) \\ \xi^*(k) & -E \end{pmatrix}$$

$$\Rightarrow \quad E^2 - |\xi(k)|^2 = 0$$

$$E_\pm(\vec{k}) = \pm|\xi(\vec{k})| =$$

$$= \pm t \left| 2\cos\frac{k_x a}{2}\, e^{-i\frac{k_y a}{2\sqrt{3}}} + e^{i\frac{k_y a}{\sqrt{3}}} \right|$$

$$= \pm t \left(2\cos\frac{k_x a}{2}\, e^{-i\frac{k_y a}{2\sqrt{3}}} + e^{i\frac{k_y a}{\sqrt{3}}}\right)^{1/2} \cdot \left(2\cos\frac{k_x a}{2}\, e^{i\frac{k_y a}{2\sqrt{3}}} + e^{-i\frac{k_y a}{\sqrt{3}}}\right)^{1/2}$$

$$= \pm t \left[\left(2\cos\frac{k_x a}{2}\right)^2 + 1 + 2\cos\frac{k_x a}{2}\, e^{-i\frac{\sqrt{3}k_y a}{2}} + e^{i\frac{\sqrt{3}k_y a}{2}}\right]^{1/2}$$

$$\frac{1}{2\sqrt{3}} + \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}}\,\frac{3}{2} = \frac{\sqrt{3}}{2}$$

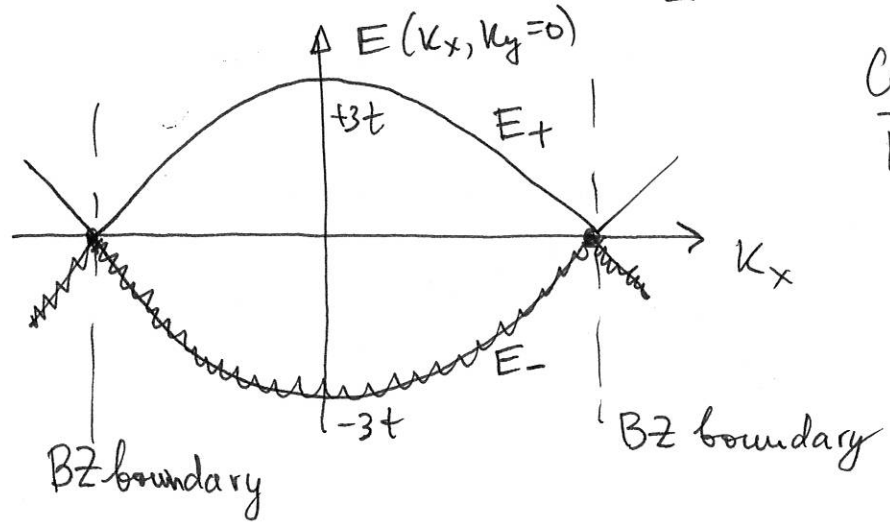
$$\underbrace{1+4\cos^2\frac{k_x a}{2}}_{2\left(1+\cos(k_x a)\right)} + 4\cos\frac{k_x a}{2}\cos\frac{\sqrt{3}}{2}k_y a = 3+2\cos(k_x a) + 4\cos\left(\frac{k_x a}{2}\right)\cos\left(\frac{\sqrt{3}}{2}k_y a\right)$$

$$\Rightarrow E_\pm(\vec{k}) = \pm t\sqrt{3+2\cos(k_x a)+4\cos\left(\frac{k_x a}{2}\right)\cos\left(\frac{\sqrt{3}}{2}k_y a\right)}$$

<u>Note</u>: 2 bands, one $E_+ \geq 0$, another $E_- \leq 0$
can touch only at points where $E=0$

By inspection: at $\vec{k} = \left(\pm\frac{4\pi}{3a}, 0\right) = \vec{K}$ or $\vec{K}'$

$$3+2\underbrace{\cos\left(\frac{4\pi}{3}\right)}_{-1/2} + 4\underbrace{\cos\left(\frac{1}{2}\,\frac{4\pi}{3}\right)}_{-1/2} = 3-1-2=0$$



Cut through $\vec{k}_y = 0$

Every C atom contributes 1 electron (out of 2 possible states)
$\Rightarrow$ the band structure should be half-filled
$\Rightarrow \begin{cases} E_-(\vec{k}) \text{ band is filled} \\ E_+(\vec{k}) \text{ is empty} \end{cases}$

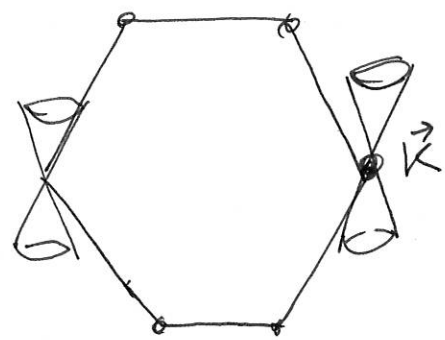
Near $\vec{K}$ ($\vec{K}'$) point: $\vec{k} = \vec{K} + \vec{q}$, $qa \ll 1$

$$E_\pm(\vec{K}+\vec{q}) = \pm t\sqrt{3 \cancel{+} 2\cos\left(\frac{4\pi}{3} + q_x a\right) + 4\cos\left(\frac{2\pi}{3} + \frac{q_x a}{2}\right)\cdot\cos\frac{\sqrt{3}q_y a}{2}}$$

$$= \pm t\,\frac{\sqrt{3}}{2}\sqrt{(q_x a)^2 + (q_y a)^2}$$

$$= \pm \hbar v |\vec{q}| + O(\vec{q}^{\,2})$$ (relativistic (Dirac spectrum))

where $v = \frac{\sqrt{3}at}{2\hbar} \approx 10^6\ m/s \ll c$



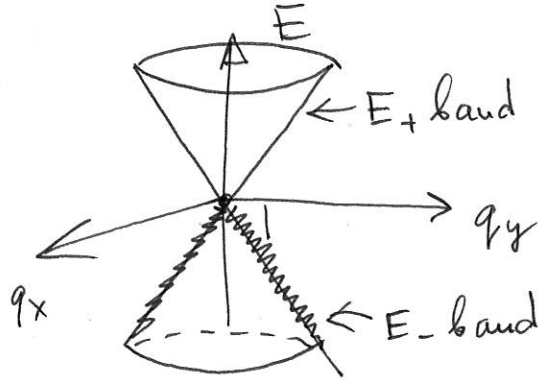
"Dirac cones" at $\vec{K}, \vec{K}'$ points of BZ

c.f. Special relativity $E = \sqrt{m^2c^4 + p^2c^2} \xrightarrow[m=0]{} cp$

$\Rightarrow$ electrons in graphene: $c \to v \approx \frac{c}{300}$

$(3\cdot 10^8\ m/s) \to (10^6\ m/s)$

Near $\vec{K}$ point:



Note: $E_\pm(\vec{k}) = E_\pm(-\vec{k})$ due to time-reversal $(t \to -t)$ symmetry $\vec{k} \to -\vec{k}$

$\left(\vec{k} = \vec{p}/\hbar \,,\ \vec{p} \sim \frac{d\vec{r}}{dt} \longrightarrow -\vec{p}\right)$ when $t \to -t$

$\Rightarrow$ if $E_\pm(\vec{k} = \vec{K}) = 0$

we immediately know that $E_\pm(\underbrace{-\vec{K}}_{\vec{K}'}) = 0$ too.

Since $\vec{K}$ and $\vec{K}' = -\vec{K}$ are not related by the reciprocal lattice vector, they represent ~~this~~ two-fold degeneracy also known as <u>valley</u> degeneracy.

Eigenfunctions: $\mathcal{H}(\vec{k})\,\psi_\lambda(\vec{k}) = E_\lambda(\vec{k}) \quad (\lambda = \pm 1)$

Let $\psi_\lambda(\vec{k}) = \begin{pmatrix} a_k^\lambda \\ b_k^\lambda \end{pmatrix}$. Note that a_k^λ (b_k^λ) represent component of wavefunction of A(B) sublattice.

$$\begin{pmatrix} 0 & \xi(k) \\ \xi^*(k) & 0 \end{pmatrix}\begin{pmatrix} a_k^\lambda \\ b_k^\lambda \end{pmatrix} = E_\lambda(k)\begin{pmatrix} a_k^\lambda \\ b_k^\lambda \end{pmatrix} = \underbrace{\lambda\,|\xi(\vec{k})|}_{E_\pm}\begin{pmatrix} a_k^\lambda \\ b_k^\lambda \end{pmatrix}$$

$$\Rightarrow \xi(k)\, b_k^\lambda = \lambda\,|\xi(k)|\, a_k^\lambda$$

$$\Rightarrow a_k^\lambda = \lambda \frac{\xi(k)}{|\xi(k)|}\, b_k^\lambda = \lambda\, e^{-i\varphi_k}\, b_k^\lambda$$

where $e^{-i\varphi_k} \equiv \dfrac{\xi(k)}{|\xi(k)|}$ and $\xi(\vec{k}) = (-t)\sum_{j=1}^{3} e^{i\vec{k}\cdot\vec{e}_j} =$

$$= (-t)\, 2\cos\left(\frac{k_x a}{2}\right) e^{-i\frac{k_y a}{2\sqrt{3}}} + e^{i\frac{k_y a}{\sqrt{3}}}$$

Normalization: $\psi_\lambda^+(\vec{k})\,\psi_\lambda(\vec{k}) = \left(a_{\lambda k}^*,\, b_{\lambda k}^*\right)\cdot\begin{pmatrix} a_{\lambda k} \\ b_{\lambda k} \end{pmatrix} = 1$

$$\Rightarrow |a_{\lambda k}|^2 + |b_{\lambda k}|^2 = 1$$

$$\Rightarrow |a_{\lambda k}| = \frac{1}{\sqrt{2}}$$

$$\psi_\lambda(\vec{k}) = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ \lambda e^{+i\varphi_k} \end{pmatrix} \quad \text{or} \quad \frac{1}{\sqrt{2}}\begin{pmatrix} e^{-i\varphi_k} \\ \lambda \end{pmatrix}$$

(differ by phase $e^{-i\varphi_k}$)

Low-energy continuum limit:

lowest energy states are located in the vicinity of $\vec{K}$ and $\vec{K}' = -\vec{K}$ points $\Rightarrow$ makes sense to derive effective low-energy Hamiltonian near $\pm\vec{K}$.

let $\vec{k} = \pm\vec{K} + \vec{q}$, $\vec{K} = \left(\frac{4\pi}{3a}, 0\right)$

↳ momentum measured from vector $(\pm\vec{K})$.

$$g(\vec{k}) = g(\pm\vec{K} + \vec{q}) = \sum_{j=1}^{3} e^{i(\pm\vec{K}+\vec{q})\cdot\vec{e}_j} = \text{(see p. 77)}$$

$$= e^{i\vec{q}\cdot\vec{e}_1} + e^{\pm i\vec{K}\cdot\vec{e}_2} e^{i\vec{q}\cdot\vec{e}_2} + e^{\pm i\vec{K}\cdot\vec{e}_3} e^{i\vec{q}\cdot\vec{e}_3}$$

$$\left(\vec{e}_1 = \frac{a}{\sqrt{3}}\vec{e}_y \Rightarrow \vec{K}\cdot\vec{e}_1 = 0.\right)$$

$$\vec{e}_2 = -\frac{a}{2}\left(\vec{e}_x + \frac{\vec{e}_y}{\sqrt{3}}\right) \Rightarrow \vec{K}\cdot\vec{e}_2 = \frac{4\pi}{3a}\left(-\frac{a}{2}\right) = -\frac{2\pi}{3}$$

$$\vec{e}_3 = \frac{a}{2}\left(\vec{e}_x - \frac{\vec{e}_y}{\sqrt{3}}\right) \qquad \vec{K}\cdot\vec{e}_3 = \frac{2\pi}{3}$$

$$g(\pm\vec{K}+\vec{q}) = e^{i\frac{q_y a}{\sqrt{3}}} + e^{\mp i\frac{2\pi}{3}} e^{-i\frac{q_x a}{2} - i\frac{q_y a}{2\sqrt{3}}}$$

$$+ e^{\pm i\frac{2\pi}{3}} e^{i\frac{q_x a}{2} - i\frac{q_y a}{2\sqrt{3}}} =$$

$$= e^{i\frac{q_y a}{\sqrt{3}}}\left(1 + e^{\mp i\frac{2\pi}{3}} e^{-i\frac{q_x a}{2} - i\frac{\sqrt{3}q_y a}{2}} + e^{\pm i\frac{2\pi}{3}} e^{i\frac{q_x a}{2} - i\frac{\sqrt{3}q_y a}{2}}\right)$$

$$\cancel{\frac{1}{\sqrt{3}}\left(1 + \frac{1}{2\sqrt{3}} \pm \frac{3}{2\sqrt{3}}\right) = \frac{\sqrt{3}}{2}}$$

$$= e^{i\frac{q_y a}{\sqrt{3}}}\left(1 + e^{\mp i\frac{2\pi}{3}}\left[1 - \frac{i}{2}(q_x a + \sqrt{3}q_y a)\right] + e^{\pm i\frac{2\pi}{3}}\left[1 - \frac{ia}{2}(-q_x + \sqrt{3}q_y)\right]\right)$$

$$= e^{i\frac{q_y a}{\sqrt{3}}}\left(1 + 2\cos\frac{2\pi}{3} - \frac{ia}{2}(q_x + \sqrt{3}q_y)e^{\mp i\frac{2\pi}{3}} - \frac{ia}{2}(-q_x + \sqrt{3}q_y)e^{\pm i\frac{2\pi}{3}}\right)$$

$$\xi(\pm\vec{K}+\vec{q}) = e^{i\frac{q_y a}{\sqrt{3}}}\left(-\frac{ia}{2}\right)\left\{q_x\left(e^{\mp i\frac{2\pi}{3}} - e^{\pm i\frac{2\pi}{3}}\right)\right.$$

$$\left. + \sqrt{3}\, q_y\left(e^{\mp i\frac{2\pi}{3}} + e^{\pm i\frac{2\pi}{3}}\right)\right\} =$$

$$= e^{i\frac{q_y a}{\sqrt{3}}}\left(-\frac{ia}{2}\right)\left(\mp 2i \underbrace{\sin\left(\frac{2\pi}{3}\right)}_{=\frac{\sqrt{3}}{2}} q_x + 2\underbrace{\cos\left(\frac{2\pi}{3}\right)}_{=-\frac{1}{2}} \sqrt{3}\, q_y\right)$$

$$= e^{i\frac{q_y a}{\sqrt{3}}}\left(-\frac{ia}{2}\right)\left(\mp i\sqrt{3}\, q_x - \sqrt{3}\, q_y\right) =$$

$$= e^{i\frac{q_y a}{\sqrt{3}}}\left(-\frac{\sqrt{3}a}{2}\right)\left(\pm q_x - i q_y\right) \cong \frac{\sqrt{3}ta}{2}\left(\pm q_x - i q_y\right)$$

insert $(-t)$ factor (p. 81)

Thus near point $\vec{K}$ (top sign in above equations)

$$\mathcal{H}_K(\vec{q}) = \begin{pmatrix} 0 & \hbar v\,(q_x - i q_y) \\ \hbar v\,(q_x + i q_y) & 0 \end{pmatrix} \qquad \boxed{v \equiv \frac{\sqrt{3}ta}{2\hbar}}$$

near $\vec{K}'$

$$\mathcal{H}_{K'}(\vec{q}) = \begin{pmatrix} 0 & \hbar v\,(-q_x - i q_y) \\ \hbar v\,(-q_x + i q_y) & 0 \end{pmatrix} = \begin{pmatrix} 0 & -v\hbar\,(q_x + i q_y) \\ -v\hbar\,(q_x - i q_y) & 0 \end{pmatrix}$$

let $\xi = \pm 1$ be <u>valley index</u> (isospin) $\xi = +1$ for $\vec{k} \approx \vec{K}$

$\xi = -1$ for $\vec{k} \approx -\vec{K} = \vec{K}'$.

$$\mathcal{H}_K(\vec{q}) = \mathcal{H}_{\xi=+1}(\vec{q}) = \hbar v\left(q_x \sigma^x + q_y \sigma^y\right)$$

$$\mathcal{H}_{K'}(\vec{q}) = \mathcal{H}_{\xi=-1}(\vec{q}) = -\hbar v\left(q_x \sigma^x - \xi q_y \sigma^y\right)$$

$$\sigma^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\xi(\vec{k}) = (-t)\left[2\cos\left(\frac{k_x a}{2}\right) e^{-i\frac{k_y a}{2\sqrt{3}}} + e^{i\frac{k_y a}{\sqrt{3}}}\right]$$

$$\vec{K} = \frac{4\pi}{3a}\hat{x}$$

$$\vec{k} = \vec{K} + \vec{q} = \left(\frac{4\pi}{3a} + q_x, q_y\right)$$

MATH $$\xi_K(\vec{q}) = (-t)\left(-\frac{\sqrt{3}a}{2}\right)(q_x - iq_y)$$

now $\vec{k} = -\vec{K} + \vec{q} \Rightarrow \xi_{K'}(\vec{q}) = (-t)\left(\frac{\sqrt{3}a}{2}\right)(q_x + iq_y)$

$$\Rightarrow \mathcal{H}_K(\vec{q}) = \frac{\sqrt{3}ta}{2}\begin{pmatrix} 0 & q_x - iq_y \\ q_x + iq_y & 0 \end{pmatrix}$$

$$\mathcal{H}_{K'}(\vec{q}) = -\frac{\sqrt{3}ta}{2}\begin{pmatrix} 0 & q_x + iq_y \\ q_x - iq_y & 0 \end{pmatrix}$$

Unified notation $\mathcal{H}_\xi(\vec{q}) = \xi \hbar v (q_x \sigma^x + \xi q_y \sigma^y)$

Eigenvalues (energy) are

$$\det\begin{pmatrix} -E & \xi\hbar v(q_x - i\xi q_y) \\ \xi\hbar v(q_x + i\xi q_y) & -E \end{pmatrix} = 0$$

$$\Rightarrow E^2 = (\xi\hbar v)^2 (q_x - i\xi q_y)(q_x + i\xi q_y) =$$

$$= (\hbar v)^2 (q_x^2 + q_y^2) = (\hbar v \vec{q})^2$$

$\Rightarrow E_\pm = \pm \hbar v |\vec{q}|$ (is independent of valley index $\xi = \pm 1$)

As before, let near $\vec{K}$ ($\xi = +1$)

$$\psi_+ = \begin{pmatrix} \psi_{A+} \\ \psi_{B+} \end{pmatrix} \text{ so that } \hbar v \begin{pmatrix} 0 & q_x - iq_y \\ q_x + iq_y & 0 \end{pmatrix} \begin{pmatrix} \psi_{A+} \\ \psi_{B+} \end{pmatrix} = E \begin{pmatrix} \psi_{A+} \\ \psi_{B+} \end{pmatrix}$$

But near $\vec{K}' = -\vec{K}$ we choose (interchange $q_x + iq_y$, $q_x - iq_y$ elements)

$$\psi_- = \begin{pmatrix} \psi_{B-} \\ \psi_{A-} \end{pmatrix} \text{ so that } \cancel{-\hbar v \begin{pmatrix} 0 & q_x - iq_y \\ q_x + iq_y & 0 \end{pmatrix} \begin{pmatrix} \psi_{B-} \\ \psi_{A-} \end{pmatrix}}$$

$$-\hbar v \begin{pmatrix} 0 & q_x - iq_y \\ q_x + iq_y & 0 \end{pmatrix} \begin{pmatrix} \psi_{B-} \\ \psi_{A-} \end{pmatrix} = E \begin{pmatrix} \psi_{B-} \\ \psi_{A-} \end{pmatrix}$$

(That is, for $\vec{k}$ near $\vec{K}'$ we interchange sublattices A and B)

Then near both points $\mathcal{H}$ can be written as

$$\mathcal{H} = \hbar v \, \tau^z \otimes \vec{\sigma} \cdot \vec{q} = \hbar v \begin{pmatrix} \vec{\sigma}\cdot\vec{q} & 0 \\ 0 & -\vec{\sigma}\cdot\vec{q} \end{pmatrix} =$$

wave function

$$\psi_K = \begin{pmatrix} \psi_{A+} \\ \psi_{B+} \\ \psi_{B-} \\ \psi_{A-} \end{pmatrix} \qquad = \hbar v \begin{pmatrix} 0 & q_x - i q_y & 0 & 0 \\ q_x + i q_y & 0 & 0 & 0 \\ 0 & 0 & 0 & -(q_x - i q_y) \\ 0 & 0 & -(q_x + i q_y) & 0 \end{pmatrix}$$

$$\tau^z \otimes \vec{\sigma} = \begin{pmatrix} \vec{\sigma} & 0 \\ 0 & -\vec{\sigma} \end{pmatrix} \qquad \left(\begin{array}{l} \vec{\tau} \text{ acts on valley space} \\ \vec{\sigma} \text{ acts on sublattice A/B space} \end{array} \right)$$

$$\vec{\sigma} \cdot \vec{q} = \sigma^x q_x + \sigma^y q_y = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} q_x + \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} q_y =$$

$$= \begin{pmatrix} 0 & q_x - i q_y \\ q_x + i q_y & 0 \end{pmatrix}$$

$$\hat{\mathcal{H}} \psi = E \psi$$

look just like 4-dimensional relativistic Dirac equation!

Near $\xi = +1$: $\hbar v \begin{pmatrix} 0 & q_x - iq_y \\ q_x + iq_y & 0 \end{pmatrix} \begin{pmatrix} \psi_A \\ \psi_B \end{pmatrix} = \lambda \hbar v |\vec{q}| \begin{pmatrix} \psi_A \\ \psi_B \end{pmatrix}$

$$(q_x - iq_y)\psi_B = \lambda |\vec{q}| \psi_A$$

$$\psi_A = \lambda \frac{q_x - iq_y}{\sqrt{q_x^2 + q_y^2}} \psi_B = \lambda \frac{|\vec{q}| e^{-i\varphi_q}}{|\vec{q}|} \psi_B$$

$$e^{-i\varphi_q} \equiv \cos\varphi_q - i \sin\varphi_q$$

$$\left.\begin{aligned} |\vec{q}| \cos\varphi_q &= q_x \\ |\vec{q}| \sin\varphi_q &= q_y \end{aligned}\right\} \Rightarrow \tan\varphi_q = \frac{q_y}{q_x}$$

$$\varphi_q = \tan^{-1}\left(\frac{q_y}{q_x}\right)$$

$$\Rightarrow \psi_{\xi=+1} = \frac{1}{\sqrt{2}} \begin{pmatrix} e^{-i\varphi_q} \\ \lambda \end{pmatrix} = \frac{e^{-i\varphi_q}}{\sqrt{2}} \begin{pmatrix} 1 \\ \lambda e^{-i\varphi_q} \end{pmatrix}.$$

Undoped graphene

$T=0$: $$N = \underset{\text{spin}}{2} \underbrace{\sum_{k\in BZ}{}'}_{} 1 = 2 \frac{V}{(2\pi)^2} \int' d^2\vec{k}$$

prime means sum over occupied states

$T\neq 0$ $$N = (2) \frac{V}{(2\pi)^2} \int_{BZ} d^2\vec{k} \left(f(E_-) + f(E_+) \right)$$

Note: no restriction on $\vec{k}$-integral except for it being over BZ

prob. to find electron with energy $E_- = -|g(k)|$

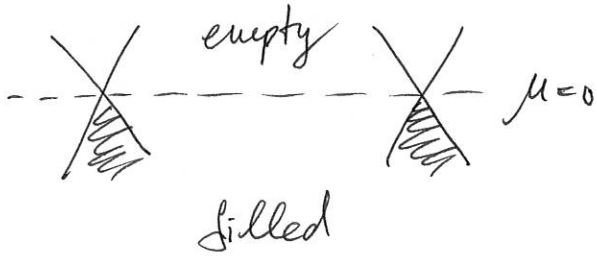
prob. to find electron with $E_+ = |g(k)|$

$$f(E_\pm) = \frac{1}{e^{\beta(E_\pm(k)-\mu)}+1} \qquad \left(\beta \equiv \frac{1}{k_B T}\right)$$

Undoped graphene at $T=0$: $E_-(\vec{k})$ is completely filled

$E_+(\vec{k})$ is empty

$\Rightarrow$ $\mu(T=0) = 0 \longrightarrow$ the system is <u>particle-hole symmetric</u>



Since $E_-(k) = -E_+(k)$, at $\mu = 0$ we have

$$f(E_-) = \frac{1}{e^{\beta E_-}+1} = \frac{1}{e^{-\beta E_+}+1} = \frac{e^{\beta E_+}}{e^{\beta E_+}+1}$$

$$\Rightarrow f(E_-) + f(E_+) = \frac{e^{\beta E_+}}{e^{\beta E_+}+1} + \frac{1}{e^{\beta E_+}+1} = 1$$

Hence $n = \frac{N}{V} = 2\int \frac{d\vec{k}}{(2\pi)^2} \underbrace{\left(f(E_+(k)) + f(E_-(k))\right)}_{1 \text{ for any } T \neq 0 \text{ as long as } \mu = 0.}$

electron density

we observe that number of particles does not change at finite T if $\mu(T) = 0 \Rightarrow$ chemical potential of undoped graphene is T-independent.

Internal energy

$$U = 2\frac{V}{(2\pi)^2}\int d\vec{k}\left(E_+(k)\, f(E_+) + E_-\, f(E_-)\right)$$

$$= 2\frac{V}{(2\pi)^2}\int d\vec{k}\left\{E_+(k)\, f(E_+) + (-E_+(k))\left(1 - f(E_+)\right)\right\}$$

$$= 2\frac{V}{(2\pi)^2}\int d\vec{k}\left\{2E_+(k)\, f(E_+) - E_+(k)\right\}$$

$$= 4\frac{V}{(2\pi)^2}\int d\vec{k}\, \frac{E_+(k)}{e^{\beta E_+(k)}+1} - 2\frac{V}{(2\pi)^2}\int d\vec{k}\, E_+(k)$$

The second integral is T-independent and represents zero-T energy of the system.

$$U(T=0) = -\frac{2V}{(2\pi)^2}\int d\vec{k}\, E_+(\vec{k})$$

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = \frac{\partial}{\partial T}\left\{\frac{4V}{(2\pi)^2}\int d\vec{k}\, \frac{E_+(k)}{e^{\beta E_+(k)}+1}\right\}$$

At low T we can restrict to contributions from the vicinity of Dirac points $\vec{K}$ and $\vec{K}' = -\vec{K}$:

near each $E_+(k) = E_+(\vec{K}+\vec{q}) = \hbar v q$

$$U(T) - U(T=0) = \frac{4V}{(2\pi)^2}\left\{\int_K d\vec{q}\, \frac{\hbar v q}{e^{\beta\hbar v q}+1} + \int_{K'} d\vec{q}\, \frac{\hbar v q}{e^{\beta\hbar v q}+1}\right\}$$

$$= \frac{8V}{(2\pi)^2}\int_0^\Lambda dq \cdot q \cdot 2\pi\, \frac{\hbar v q}{e^{\hbar v q/T}+1} = \left(x = \frac{\hbar v q}{T}\right)$$

$$= \frac{8V}{2\pi}\left(\frac{T}{\hbar v}\right)^3 \int_0^{\hbar v\Lambda/T} dx\, x\, \frac{\hbar v x}{e^x+1} =$$

$$= \frac{4V}{\pi} k_B T \left(\frac{k_B T}{\hbar v}\right)^2 \int_0^\infty dx\, \frac{x^2}{e^x+1} \Rightarrow C_V \sim V k_B \left(\frac{k_B T}{\hbar v}\right)^2.$$

For usual metals $C_V \sim T$ of course.

The integral is convergent $\Rightarrow$ replace cutoff $\Lambda \to \infty$

$$\int_0^\infty dx\, \frac{x^2}{e^x+1} = \frac{3}{2}\zeta(3) = \frac{3}{2}\,\mathrm{Zeta}(3) = \frac{3}{2}\sum_{n=1}^\infty \frac{1}{n^3}$$

Density of states:

$$g(E) = \sum_{\varepsilon_i} \delta(E-\varepsilon_i) = \overset{\text{spin}}{2} \sum_k \left(\delta(E-\varepsilon_+(k)) + \delta(E-\varepsilon_-(k))\right)$$

$$= 2\cdot \overset{\text{valley}}{2} \sum_k \left(\delta(E-\hbar v|k|) + \delta(E+\hbar v|k|)\right)$$

$$= 4\,\frac{V}{(2\pi)^2} \underbrace{\int d\vec{k}}_{\int dk\cdot k\cdot 2\pi}\ \delta(E-\hbar v k)$$

let $E>0$: $g(E) = \frac{4V}{2\pi}\int_0^\infty dk\, k\, \delta(E-\hbar v k) =$

$$= \left(k=\frac{x}{\hbar v}\right) \quad = \frac{4V}{2\pi}\int_0^\infty \frac{dx}{\hbar v}\,\frac{x}{\hbar v}\,\delta(E-x) = \frac{2V}{\pi}\,\frac{E}{(\hbar v)^2}$$

let $E<0$: $g(E) = \frac{4V}{(2\pi)^2}\int_0^\infty dk\cdot k\cdot 2\pi\ \delta(E+\hbar v k)$

$$= \frac{2V}{\pi}\int_0^\infty \frac{dx\cdot x}{(\hbar v)^2}\,\delta(E+x) = \frac{2V}{\pi}\,\frac{-E}{(\hbar v)^2}$$

$\Rightarrow$ for any sign of E

$$g(E) = \frac{2V}{\pi}\,\frac{|E|}{\hbar^2 v^2}.$$